Abdelrahman Osheba

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Abde1rahman-Osheba

Summary

An ambitious and progressive student of AI with a demonstrated capacity for taking intelligent systems and the academy's knowledge from the classroom or theory into the real world and into practice in society. Proven experience in developing applied, computational solutions that create impact and value in healthcare systems and research programs, merging leading-edge thinking with practical, real-world applications for those experiencing the impact. Acknowledged for excellence in scholarship, leadership, and mentoring in the AI community. Certified in leading technologies and cloud platforms..

Education

Galala University *Sept 2022 – Present*
BS in Computer Science and Engineering

- GPA: 3.856/4.0
- **Program:** Artificial Intelligence Science expected to graduate by 2026

Experience

Technical Subjects and Mathematics Instructor *Italian High School for Science and Technology (IHSS)* *New Cairo, EG*
Aug 2025 – Present

- Deliver practical and theoretical instructional sessions in C++ programming, competitive programming, and applied computing projects.
- Design and lead hands-on field training focused on programming implementation, debugging, and project development workflows.
- Develop full teaching materials, including course syllabus, slides, curriculum, coding exercises, assessments, quizzes, and exams for C++ and Computer Networks.
- Mentor students in building individual and collaborative software projects, guiding them through problem-solving, algorithmic thinking, and optimization strategies.
- Provide mathematics instruction and concept clarification.
- Support evaluation and continuous improvement of technical learning outcomes through structured assessments and performance tracking.

Intern *Itida and NTI* *Remote*
Aug 2025 – Sep 2025

- Developed NLP applications with a focus on large language models (LLMs) and conversational agents.
- Gained hands-on experience in fine-tuning LLMs, semantic analysis, and building end-to-end AI-driven dialogue systems.

Mobile Networks Intern *Nokia* *Maadi, EG*
July 2025 – Aug 2025

- Gained hands-on experience in mobile network technologies, digital twin modeling, and applied AI for resilience and optimization.
- Contributed to NOVA Bot, a GNS3-based digital twin for network technologies simulating operator topologies, routing (OSPF/BGP/MPLS), and performance metrics for AI driven decision support.

Trainee *TachyHealth* *Remote*
Oct 2024 – Dec 2024

- Gained knowledge in claim validation, data visualization, and software development, fundamental concepts of the healthcare ecosystem, including AI, machine learning, and claim management.
- Conducted a comprehensive research and implementation study on the design and deployment of intelligent healthcare data management systems to enhance patient data handling, security, and predictive analytics

within medical institutions.

Publications

Leveraging Large Language Models for Smart Pharmacy Systems: Enhancing Drug Safety and Operational Efficiency Feb 2025

Abdelrahman Osheba, Ahmed Abou-Elela Osama Adel Youssef Moaad Tamer Abuhmed Shaker El-Sappagh

10.1109/IMCOM64595.2025.10857582

Projects

Smart Pharmacy Using Large Language Models

- Developed an AI-powered smart pharmacy system leveraging Large Language Models (LLMs) to automate drug interaction detection, enhance operational efficiency, and improve patient safety.
- Tools Used: Langchain, Python, Whisper and Llama

CardAI

- Developed an AI-powered heart sound analysis system with multi-level feature encoding for accurate classification of cardiac conditions. Real-time data transmission to a pre-trained AI model, improving diagnostic precision.
- Tools Used: Python, Pyaudio and RNNs

LLM Healthcare with RAG AI Agent for Medical query answering

- Developed an AI healthcare assistant integrating Retrieval-Augmented Generation (RAG) with FAISS memory and fallback web automation, enabling reliable, real-time medical query answering with conversational memory.
- Tools Used: Python, Langchain, LangGraph, Crewai, RAG and FAISS

NeoMedica - Medical Knowledge Graph System

Developed an intelligent medical advisor using Neo4j, Streamlit, and Python to model diseases, symptoms, and treatments as an interactive knowledge graph, enabling diagnostic support and medical education through real-time query and visualization.

Tools Used: Python, Streamlit, LangGraph, RAG and Neo4j

- **COVID-19 Clinical Data Analysis (Sírio-Libanês Hospital, Brazil)**

  Conducted exploratory data analysis (EDA) on patient-level clinical records to identify key risk factors and trends in COVID-19 severity.

  Tools Used: Python and sklearn

- **Adaptation of the Sand Cat Search Optimization (SCSO) algorithm to solve the Traveling Salesman Problem (TSP)**

  Using python in building and optimizing Sand Cat Search Optimization algorithm in solving popular problem as Traveling Salesman Problem.

  Tools Used: Python

- **Quantum Image Classification with QNNs (Qiskit)**

  A project where I built and trained quantum neural networks using Qiskit to classify simple 2×4-pixel images into horizontal vs. vertical line patterns, exploring feature maps, entanglement strategies, and variational ansatz design.

  Tools Used: Python and Qsikit

Competitions and Participations

- **The 2023 ICPC ECPC Qualifications Collegiate Programming Contest** *August 2023*
  The 2024 ICPC ECPC Qualifications Collegiate Programming Contest *July 2024*
  3rd Artificial Intelligence Olympiad – 2nd Place Winner *March 2024*

AstraZeneca & Faculty of Pharmacy, Galala University Competition *Innovation and Recent Trends in Healthcare – 1st Place Winner* — *April 2024*

Faculty of Medicine Galala University *Antimicrobial Stewardship: From Policy to Practice - 2nd place winner* — *December 2025*

Certificates

Introduction to Artificial Intelligence and Applications *Zewail City & Impact* — *August 2023*

AWS Academy Cloud Foundations *AWS* — *December 2023*

TOT Python for Kids *SKAYZ-STEM* — *January 2025*

AWS Academy Machine Learning Foundations *AWS* — *May 2025*

Huawei Big Data Associate *NTI / Huawei Egyptian Talent Academy* — *July 2025*

Introducing Generative AI with AWS *Udacity & AWS* — *July 2025*

Machine Learning Specialization *Stanford Online & DeepLearning.AI* — *August 2025*

Quantum Machine Learning *IBM* — *August 2025*

Volunteering Work

CAISAIS Organizers Committee *IEEE GUSB – Galala University* — *3rd–5th September 2023*

Techne Drifts Logistics Team *Galala University* — *February 2024*

CSDGTAI Organizers Committee *IEEE GUSB – Galala University* — *9th–11th December 2024*

Extracurricular Activities

IEEE GUSB IT Member — *Sept 2022 – Nov 2023*

IEEE GUSB IT Head — *Nov 2023 – Oct 2024*

GDSC GU Project Management Member — *May 2024 – Aug 2024*

GDG GU Guest Lecturer — *Aug 2024 – May 2025*

ICPC GU Community Programming Trainer — *Aug 2024 – Aug 2025*

IEEE GUSB Executive Secretary — *Aug 2024 – Aug 2025*

GDG GU Technical Mentor — *Oct 2024 – Dec 2024*

Technologies

Languages: Python, C++, SQL

Frameworks & Tools: PyTorch, TensorFlow, Scikit-learn, Transformers, Streamlit, Neo4j, Qiskit, FAISS, GNS3

Platforms & Cloud: AWS, Microsoft SQL Server, Huawei